Problem No: 01
Problem Name: Study and Simulation of the 3D Radiation Pattern of a Dipole Antenna

Objective

- To understand the basic structure and working principle of a dipole antenna.
- To analyze the mathematical model of the dipole radiation pattern.
- To simulate and visualize both 2D and 3D radiation patterns using Python.

Theory

A dipole antenna is one of the simplest and most widely used antennas in wireless communication systems. It consists of two identical conductive elements and operates most efficiently when its total length is approximately half of the wavelength ($\lambda/2$), commonly known as a half-wave dipole antenna. Due to its simple structure, ease of fabrication, and reliable performance, it is extensively used in radio, television broadcasting, and communication systems.

The radiation pattern of an antenna describes how electromagnetic energy is distributed in space. It indicates the directions of maximum and minimum radiation. For a half-wave dipole antenna, the radiated electric field intensity varies with the elevation angle θ and is given by:

$$E(\theta) \propto \sin(\theta)$$

After normalization, the radiation pattern can be expressed as:

$$R(\theta) = |\sin(\theta)|$$

This equation shows that the radiation is maximum at $\theta = 90^\circ$, meaning the energy is strongest in the direction perpendicular to the antenna axis. Conversely, the radiation becomes zero at $\theta = 0^\circ$ and $\theta = 180^\circ$, indicating no radiation along the axis of the antenna. As a result, the dipole antenna produces a characteristic figure-eight radiation pattern in the 2D plane and behaves as an omnidirectional radiator in the horizontal plane.

In practical antenna systems, several important parameters are used to describe performance. One such parameter is directivity, which measures how concentrated the radiation is in a particular direction. A half-wave dipole antenna has a typical directivity of approximately 2.15 dBi. Another important parameter is radiation resistance, which represents the portion of antenna resistance responsible for radiation of electromagnetic energy.

For three-dimensional visualization, the radiation pattern is initially defined in spherical coordinates and then converted into Cartesian coordinates. This transformation enables easier graphical representation of the radiation pattern in 3D space. The coordinate transformation is given by:

$$X = R \sin\theta \cos\phi$$

$$Y = R \sin\theta \sin\phi$$

$$Z = R \cos\theta$$

The resulting 3D radiation pattern typically forms a toroidal (doughnut-shaped) structure, indicating that radiation is maximum in the plane perpendicular to the antenna and minimum along its axis.

Thus, both theoretical analysis and simulation play a crucial role in understanding the spatial distribution and directional characteristics of electromagnetic radiation from a dipole antenna, which is fundamental in antenna design and communication systems.

Source Code

```
# Dipole Antenna-3D
import numpy as np
import matplotlib.pyplot as plt

# Angle values
theta = np.linspace(0, np.pi, 100)
phi = np.linspace(0, 2*np.pi, 100)
theta, phi = np.meshgrid(theta, phi)
# Radiation pattern
R = np.abs(np.sin(theta))
# Coordinate conversion
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

# Plot
fig = plt.figure()
ax = fig.add_subplot(projection='3d')
ax.plot_surface(X, Y, Z)
ax.set_title("3D Radiation Pattern of Dipole Antenna")
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")
plt.show()
```

Output

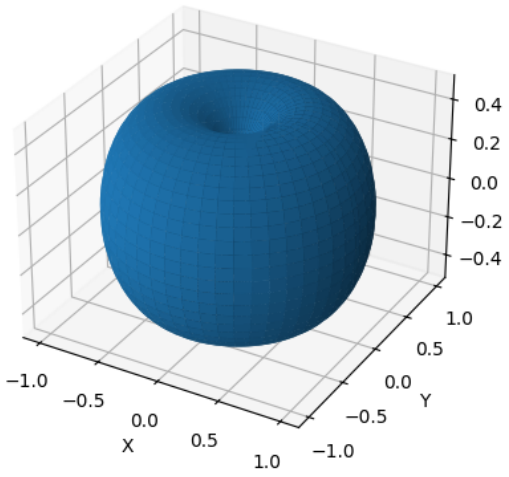


Figure 1: 3D Radiation Patterns of Dipole Antenna

Result and Discussion

The simulation results clearly demonstrate the radiation characteristics of a half-wave dipole antenna. The 2D radiation pattern exhibits a figure-eight shape, indicating that maximum radiation occurs perpendicular to the antenna axis, while there is negligible radiation along the axis itself. The 3D radiation pattern forms a toroidal or doughnut-shaped structure, which confirms that the antenna radiates uniformly in the horizontal plane but not in the vertical direction. These results are consistent with the theoretical expression $R(\theta) = |\sin(\theta)|$, validating the correctness of the simulation. Overall, the experiment successfully visualizes how electromagnetic energy is distributed in space for a dipole antenna.

Problem No: 02
Problem Name: Study and Simulation of the 3D Radiation Pattern of an Isotropic Antenna

Objective

- To understand the concept of an isotropic antenna.
- To study uniform radiation characteristics in all directions.
- To analyze the ideal radiation model mathematically.
- To simulate and visualize 2D and 3D radiation patterns using Python.

Theory

An isotropic antenna is an ideal and hypothetical antenna that radiates electromagnetic energy equally in all directions in three-dimensional space. Although such an antenna cannot be physically realized, it plays a fundamental role in antenna theory as a reference model for evaluating and comparing the performance of practical antennas.

Unlike real antennas such as dipole, horn, or patch antennas, an isotropic antenna does not exhibit any directional preference. This means that the radiation intensity remains constant regardless of the observation angle. Due to this unique property, it is widely used as a standard reference for defining antenna parameters such as gain and directivity.

Mathematically, the radiation pattern of an isotropic antenna is expressed as:

$$R(\theta, \phi) = \text{constant}$$

This equation indicates that the radiated power is uniformly distributed in all directions, resulting in a perfect spherical radiation pattern in three-dimensional space.

In a two-dimensional representation, the radiation pattern appears as a perfect circle because the magnitude of radiation is identical for all values of θ. This can be written as:

$$R(\theta) = 1$$

This uniform distribution implies that there are no nulls, side lobes, or preferred directions in the radiation pattern, making it completely omnidirectional.

For visualization purposes, the radiation pattern defined in spherical coordinates is converted into Cartesian coordinates using the following relations:

$$X = R \sin\theta \cos\phi$$

$$Y = R \sin\theta \sin\phi$$

$$Z = R \cos\theta$$

Since R is constant, these equations generate a perfect sphere in three-dimensional space, clearly illustrating uniform radiation in all directions.

The isotropic antenna has a theoretical gain of 1 (or 0 dBi), which serves as a baseline for comparing other antennas. For instance, if an antenna has a gain of 3 dBi, it means that it radiates 3 dB more power in a particular direction compared to an isotropic antenna.

Although isotropic antennas are not physically realizable, they are extremely important in understanding antenna behavior and performance metrics such as gain, directivity, and efficiency. Therefore, the concept of an isotropic antenna forms the foundation for the analysis and design of practical antenna systems.

Source Code

```
# Isotropic Antenna-3D
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi, 200)
phi = np.linspace(0, 2*np.pi, 200)
theta, phi = np.meshgrid(theta, phi)
R = np.abs(np.cos(theta))
X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

ax = plt.figure().add_subplot(projection='3d')
ax.plot_surface(X, Y, Z, cmap="plasma")
ax.set_title("Isotropic Antenna (3D)")
plt.show()
```

Output

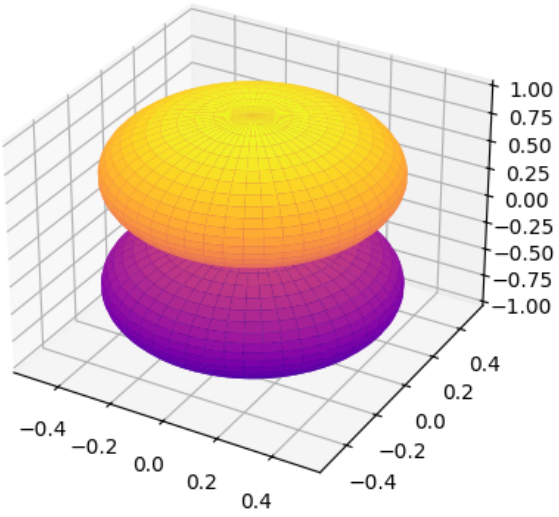


Figure 2: 3D Radiation Patterns of Isotropic Antenna

Result and Discussion

The simulation results clearly illustrate the radiation characteristics of an isotropic antenna. The 2D radiation pattern forms a perfect circle, indicating that the radiation intensity is equal in all directions. In the 3D visualization, the radiation pattern resembles a symmetrical structure, demonstrating uniform energy distribution in space. Although the practical implementation of an isotropic antenna is not possible, it serves as an ideal reference model. The results confirm that the radiation remains constant irrespective of direction, which aligns with the theoretical concept of isotropic radiation. Therefore, the simulation successfully represents the ideal behavior of an isotropic antenna.

Problem No: 03
Problem Name: Study and Simulation of the 3D Radiation Pattern of a Horn Antenna

Objective

- To understand the working principle of a horn antenna.
- To study the directional radiation characteristics of horn antennas.
- To analyze the beam pattern and directivity.
- To simulate and visualize 2D and 3D radiation patterns using Python.

Theory

A horn antenna is a highly directional antenna that operates efficiently in the microwave frequency range. It is formed by gradually flaring the end of a waveguide, which helps in reducing reflections and improving impedance matching between the waveguide and free space. This design allows electromagnetic waves to radiate efficiently with minimal loss.

Unlike dipole or isotropic antennas, which radiate energy in multiple directions, a horn antenna focuses energy in a specific direction. As a result, it exhibits high directivity and gain. The radiation pattern of a horn antenna is characterized by a dominant main lobe in the forward direction and relatively small side lobes, indicating that most of the radiated power is concentrated along a particular axis.

The radiation behavior of a horn antenna can be approximated using a cosine-based function:

$$R(\theta) = |\cos(\theta)|^n$$

where n is a constant that controls the beamwidth of the radiation pattern. A higher value of n results in a narrower main lobe, which indicates increased directivity and better beam focusing.

In the 2D radiation pattern, the horn antenna produces a single strong main lobe pointing in one direction, demonstrating that the radiation is highly directional. The absence of significant radiation in other directions confirms its efficiency in focusing electromagnetic energy.

For three-dimensional visualization, the radiation pattern is first expressed in spherical coordinates and then converted into Cartesian coordinates using the following relations:

$$X = R \sin\theta \cos\phi$$

$$Y = R \sin\theta \sin\phi$$

$$Z = R \cos\theta$$

The resulting 3D radiation pattern forms a highly concentrated beam along one axis, showing a sharp and focused energy distribution. This directional property makes horn antennas particularly suitable for applications such as radar systems, satellite communication, microwave links, and antenna measurements.

Additionally, horn antennas are widely used as standard gain antennas because of their predictable radiation characteristics and ease of analysis. Thus, the study of horn antennas provides valuable insight into high-frequency directional radiation and antenna design principles.

Source Code

```
# Horn Antenna-3D
import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, np.pi, 250)
phi = np.linspace(0, 2*np.pi, 250)
theta, phi = np.meshgrid(theta, phi)
```

```
n = 6
R = np.abs(np.cos(theta))**n

X = R * np.sin(theta) * np.cos(phi)
Y = R * np.sin(theta) * np.sin(phi)
Z = R * np.cos(theta)

ax = plt.figure().add_subplot(projection='3d')
ax.plot_surface(X, Y, Z, cmap="inferno")
ax.set_title("Horn Antenna (3D)")
plt.show()
```

Output

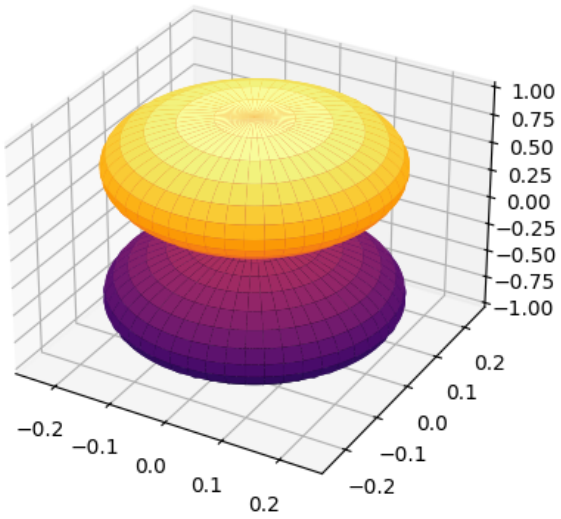


Figure 3: 3D Radiation Patterns of Horn Antenna

Result and Discussion

The simulation results clearly demonstrate the directional radiation characteristics of a horn antenna. The 2D radiation pattern shows a strong main lobe concentrated in a specific direction, indicating that most of the energy is radiated forward. Unlike isotropic and dipole antennas, the horn antenna does not radiate equally in all directions. The 3D radiation pattern further confirms this behavior by forming a focused beam along one axis. This high directivity makes horn antennas highly suitable for applications such as radar, satellite communication, and microwave transmission. The simulation results are consistent with theoretical expectations, validating the effectiveness of the model.

Problem No: 04
Problem Name: Study and Simulation of the 2D Radiation Pattern of a Linear Antenna Array

Objective

- To understand the concept of a linear antenna array.
- To study the effect of multiple antenna elements on radiation pattern.
- To analyze the formation of main lobe and side lobes.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A linear antenna array is formed by arranging multiple identical antenna elements in a straight line with uniform spacing. By properly controlling the amplitude and phase of excitation of each element, the overall radiation pattern can be shaped to achieve higher directivity, improved beam steering, and desired radiation characteristics. This makes antenna arrays highly useful in modern communication and radar systems.

The total radiation pattern of a linear array is determined by the product of the individual element pattern and the array factor (AF). The array factor represents the combined effect of all antenna elements and plays a crucial role in determining both the direction and shape of radiation.

For a uniform linear array with N elements and inter-element spacing d, the array factor is given by:

$$AF(\theta) = \left| \sum_{n=0}^{N-1} e^{jn(kd\cos\theta+\beta)} \right|$$

where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- λ is the wavelength
- d is the spacing between adjacent elements
- β is the progressive phase shift between elements

The phase difference β determines the direction of maximum radiation. When $\beta = 0$, the array operates in broadside mode, where the maximum radiation occurs perpendicular to the array axis ($\theta = 90^\circ$). By adjusting β, the main beam can be steered in different directions without physically moving the antenna, a technique known as beam steering.

The radiation pattern of a linear array typically consists of a main lobe, which indicates the direction of maximum radiation, and several side lobes, which represent unwanted radiation in other directions. The angular width of the main lobe, known as beamwidth, decreases as the number of elements increases, resulting in higher directivity and better focusing of electromagnetic energy.

The spacing between elements also significantly affects the radiation pattern. If the spacing d is too large, unwanted grating lobes may appear, leading to multiple main beams and reduced performance. To avoid this, a common design choice is $d = \lambda/2$, which ensures efficient radiation without grating lobes.

Source Code

```
# Linear Antenna Array (2D)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
N = 8               # Number of elements
d = 0.5             # Spacing (lambda/2)
```

```
k = 2 * np.pi       # Wave number
beta = 0            # Phase difference
theta = np.linspace(0, 2*np.pi, 1000)
# Array Factor
AF = np.zeros_like(theta, dtype=complex)
for n in range(N):
    AF += np.exp(1j * n * (k * d * np.cos(theta) + beta))
AF = np.abs(AF)
AF = AF / np.max(AF)    # Normalize

# Plot
ax = plt.subplot(111, projection="polar")
ax.plot(theta, AF)

ax.set_title("Linear Antenna Array (2D)")
plt.show()
```

Output

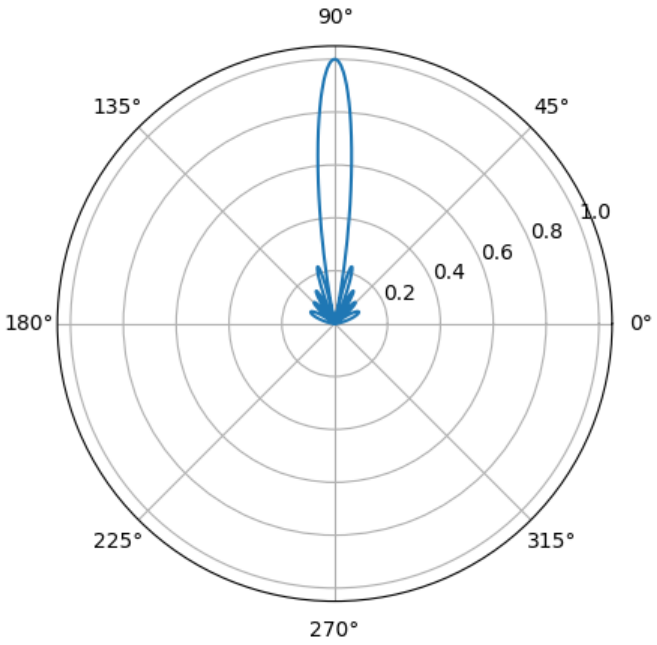


Figure 4: 2D Radiation Pattern of Linear Antenna Array

Result and Discussion

The simulation results demonstrate the radiation characteristics of a linear antenna array. The 2D radiation pattern clearly shows a strong and narrow main lobe, indicating that most of the energy is concentrated in a specific direction. Additionally, several smaller side lobes are observed, which represent secondary radiation in other directions. Compared to a single antenna element, the linear array provides significantly higher directivity and improved beam control. The presence of side lobes indicates that some energy is still distributed outside the main beam, but overall, the array configuration enhances the efficiency of directional radiation. The results are consistent with theoretical expectations, confirming the effectiveness of the array factor model.

Problem No: 05
Problem Name: Study and Simulation of the 2D Radiation Pattern of a Broadside 3-Element Array Antenna

Objective

- To understand the concept of a broadside array antenna.
- To study the radiation characteristics of a 3-element antenna array.
- To analyze the formation of main lobe and side lobes.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A broadside array antenna is a type of linear antenna array in which multiple identical antenna elements are arranged in a straight line with equal spacing, and all elements are excited with equal amplitude and phase. In this configuration, the maximum radiation occurs in a direction perpendicular to the axis of the array, which is known as the broadside direction.

The overall radiation pattern of an array antenna depends on both the individual element pattern and the array factor (AF). The array factor represents the combined effect of all antenna elements and plays a crucial role in shaping the radiation pattern.

For a uniform linear array with N elements and inter-element spacing d, the array factor is given by:

$$AF(\theta) = \left| \sum_{n=0}^{N-1} e^{jn(kd\cos\theta+\beta)} \right|$$

For a broadside array, the phase difference $\beta = 0$, meaning all elements radiate in phase. This results in constructive interference in the direction $\theta = 90°$, producing a strong main lobe perpendicular to the array axis.

In the case of a 3-element broadside array, the radiation pattern exhibits a dominant main lobe centered at $\theta = 90°$ along with smaller side lobes on either side. The presence of multiple elements increases the directivity of the antenna, allowing energy to be concentrated in a specific direction.

The beamwidth of the main lobe decreases as the number of elements increases, leading to a sharper and more focused radiation pattern. However, side lobes are also formed due to interference between waves from different elements, which represent unwanted radiation in other directions.

The 2D radiation pattern of a broadside array typically shows symmetry about the broadside direction, confirming that the antenna radiates equally on both sides of the perpendicular axis. This property makes broadside arrays highly useful in applications such as radar systems, wireless communication, and broadcasting, where controlled directional radiation is required.

Thus, the study of a 3-element broadside array helps in understanding how multiple antenna elements interact to shape the radiation pattern and improve antenna performance.

Source Code

Step 1: Broadside 3-Element Array (2D Radiation Pattern)

```
# Broadside array antenna (3-element)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 3   # number of elements
# Angle range
```

```
theta = np.linspace(0, np.pi, 1000)
# Array Factor
AF = np.zeros_like(theta, dtype=complex)
for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))
# Normalize
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

# Plot
plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)

ax.set_title("2D Radiation Pattern (3-Element Broadside Array)")
plt.show()
```

Output

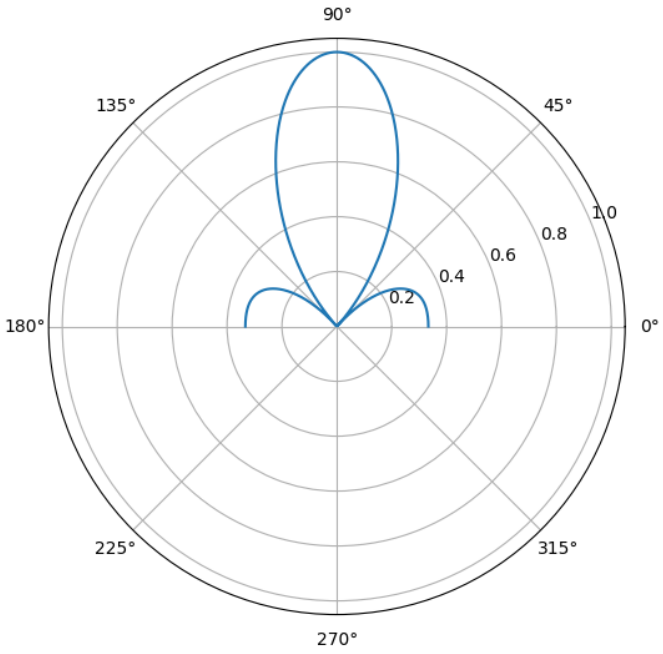


Figure 5: 2D Radiation Pattern of 3-Element Broadside Array Antenna

Result and Discussion

The simulation results illustrate the radiation characteristics of a 3-element broadside array antenna. The 2D radiation pattern shows a strong main lobe directed perpendicular to the array axis, confirming the broadside nature of the antenna. Additionally, smaller side lobes are observed on either side of the main beam, indicating partial radiation in other directions. Compared to a single antenna element, the array significantly improves directivity and beam focusing. However, the presence of side lobes suggests that some power is not confined entirely to the main direction. Overall, the results align well with theoretical expectations and demonstrate the effectiveness of broadside array configuration in achieving directional radiation.

Problem No: 06
Problem Name: Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Phase ($d = \lambda/2$)

আমরা দেখতে চাই—দুটি antenna একই সময়ে সমান শক্তিতে এবং একই phase-এ signal পাঠালে চারদিকে signal-এর radiation কেমন হয়।

Objective

- To understand the concept of interference between two isotropic sources.
- To study the far-field radiation pattern of two-element antenna systems.
- To analyze the effect of equal amplitude and phase on radiation pattern.
- To simulate and visualize the radiation pattern using Python.

Theory

In antenna theory, when two isotropic point sources radiate electromagnetic waves simultaneously, the resulting radiation pattern is determined by the principle of superposition. If the sources have equal amplitude and are in phase, their waves interfere constructively and destructively at different angles, producing a characteristic interference pattern.

For two isotropic sources separated by a distance d, the array factor (AF) in the far-field region is given by:

$$AF(\theta) = 2\cos\left(\frac{kd}{2}\cos\theta\right)$$

where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- λ is the wavelength
- d is the distance between the two sources

When the spacing is $d = \lambda/2$, the expression becomes:

$$AF(\theta) = 2\cos\left(\frac{\pi}{2}\cos\theta\right)$$

This equation shows that the radiation pattern depends on the cosine of the angle, leading to alternating regions of constructive and destructive interference.

Constructive interference occurs when the waves from both sources arrive in phase, resulting in maximum radiation (main lobes). Destructive interference occurs when the waves cancel each other, producing nulls in the radiation pattern.

The far-field radiation pattern is symmetric and typically exhibits two main lobes and several nulls depending on the spacing and wavelength. This pattern demonstrates how multiple sources can shape and control radiation characteristics.

Source Code

```
# Two Isotropic Point Sources (Equal Amplitude & Phase)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam
# Angle range
theta = np.linspace(0, 2*np.pi, 1000)
```

```
# Array Factor
AF = 2 * np.cos((k * d / 2) * np.cos(theta))
# Normalize
AF_norm = np.abs(AF) / np.max(np.abs(AF))

# Plot
plt.figure(figsize=(6,6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nTwo Isotropic Point Sources\nEqual Amplitude &
    Phase (d =   /2)")
plt.show()
```

Output

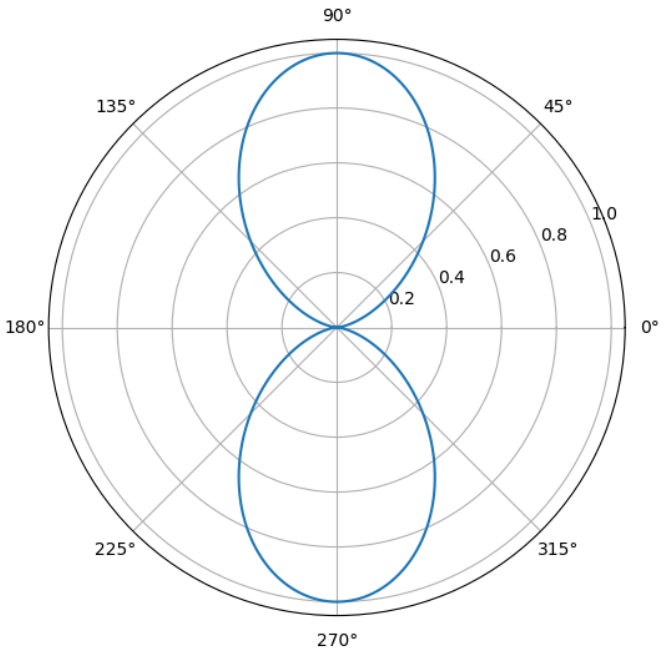


Figure 6: Far-Field Radiation Pattern of Two Isotropic Point Sources

Result and Discussion

The simulation results illustrate the far-field radiation pattern produced by two isotropic point sources with equal amplitude and phase. The pattern shows clear regions of constructive and destructive interference, resulting in distinct lobes and nulls. The maximum radiation occurs where the waves from both sources combine constructively, while minimum radiation occurs due to destructive interference. The symmetric nature of the pattern confirms that both sources are identical and equally excited. This experiment demonstrates how multiple sources can interact to shape the radiation pattern, which is a fundamental concept in antenna array design. The results are consistent with theoretical predictions, validating the correctness of the model.

Problem No: 07
Problem Name: Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Equal Amplitude and Opposite Phase ($d = \lambda/2$)

Objective

- To understand the effect of phase difference between two sources.
- To study the far-field radiation pattern for opposite phase excitation.
- To analyze destructive and constructive interference.
- To simulate and visualize the radiation pattern using Python.

Theory

When two isotropic point sources radiate electromagnetic waves with equal amplitude but opposite phase (i.e., a phase difference of π radians), the resulting radiation pattern is significantly altered due to interference effects. According to the principle of superposition, the total field is the sum of the individual fields from both sources.

For two sources separated by a distance d, the array factor (AF) considering a phase difference β is given by:

$$AF(\theta) = 2\cos\left(\frac{kd}{2}\cos\theta + \frac{\beta}{2}\right)$$

where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- β is the phase difference between sources

For opposite phase excitation, $\beta = \pi$, so the expression becomes:

$$AF(\theta) = 2\cos\left(\frac{kd}{2}\cos\theta + \frac{\pi}{2}\right)$$

Using trigonometric identity, this simplifies to:

$$AF(\theta) = 2\sin\left(\frac{kd}{2}\cos\theta\right)$$

For spacing $d = \lambda/2$, we get:

$$AF(\theta) = 2\sin\left(\frac{\pi}{2}\cos\theta\right)$$

This equation indicates that the radiation pattern will have nulls where the sine function becomes zero and maxima where it reaches its peak.

Unlike the equal-phase case, the radiation pattern here shows minimum radiation in the broadside direction ($\theta = 90°$) and maximum radiation along the axis ($\theta = 0°$ and $180°$). This results in a pattern similar to an end-fire array.

Thus, phase difference plays a crucial role in controlling the direction of radiation and shaping the antenna pattern.

Source Code

Step 1: Two Isotropic Sources (Opposite Phase)

```
# Two Isotropic Point Sources (Opposite Phase)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam
# Angle range
theta = np.linspace(0, 2*np.pi, 1000)
# Array Factor (Opposite Phase)
AF = 2 * np.sin((k * d / 2) * np.cos(theta))
# Normalize
AF_norm = np.abs(AF) / np.max(np.abs(AF))

# Plot
plt.figure(figsize=(6,6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nTwo Isotropic Sources (Opposite Phase)\n(d =   /2)"
    )
plt.show()
```

Output

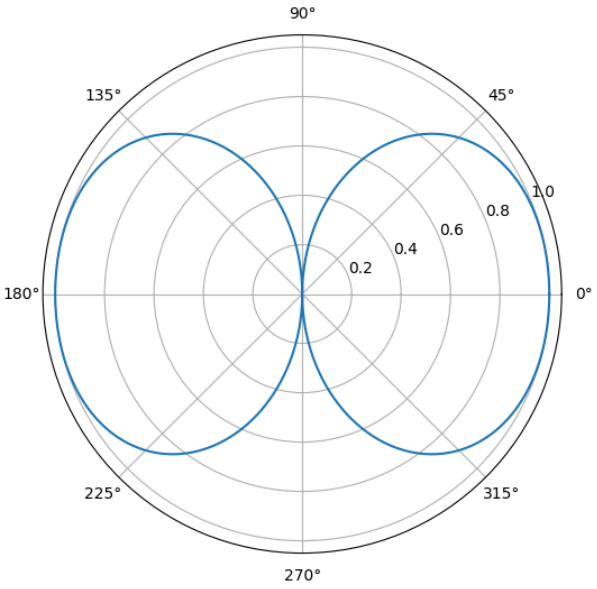


Figure 7: Far-Field Radiation Pattern of Two Isotropic Sources (Opposite Phase)

Result and Discussion

The simulation results show the far-field radiation pattern of two isotropic point sources with equal amplitude and opposite phase. The pattern clearly differs from the equal-phase case, as the maximum radiation now occurs along the axis of the sources ($\theta = 0°$ and $180°$), while a null is observed in the broadside direction ($\theta = 90°$). This behavior is due to destructive interference at the perpendicular direction and constructive interference along the axis. The pattern resembles an end-fire radiation pattern, demonstrating how phase difference can significantly alter the direction of radiation. The results are consistent with theoretical expectations and highlight the importance of phase control in antenna array design.

Problem No: 08
Problem Name: Study and Plot the Far-Field Radiation Pattern of Two Isotropic Point Sources with Unequal Amplitude and Opposite Phase ($d = \lambda/2$)

Objective

- To understand the effect of unequal amplitude in antenna arrays.
- To study the radiation pattern for opposite phase excitation.
- To analyze how amplitude variation affects side lobes and nulls.
- To simulate and visualize the radiation pattern using Python.

Theory

When two isotropic point sources radiate electromagnetic waves with unequal amplitudes and opposite phase, the resulting radiation pattern is governed by both amplitude weighting and phase difference. According to the principle of superposition, the total radiated field at any observation point is the vector sum of the individual fields produced by each source.

For two isotropic sources with amplitudes A_1 and A_2, separated by a distance d, and having a phase difference β, the array factor (AF) can be expressed as:

$$AF(\theta) = A_1 + A_2 e^{j(kd\cos\theta+\beta)}$$

where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- λ is the wavelength
- d is the spacing between the sources
- β is the phase difference between the sources

For opposite phase excitation, the phase difference is $\beta = \pi$, and the expression becomes:

$$AF(\theta) = A_1 - A_2 e^{j(kd\cos\theta)}$$

This indicates that the signals from the two sources are out of phase, leading to destructive interference in certain directions and constructive interference in others.

When the amplitudes are unequal ($A_1 \neq A_2$), the interference pattern becomes asymmetric. Unlike the equal amplitude case, the nulls are no longer completely zero because one source contributes more energy than the other. As a result, the radiation pattern is distorted, and the maxima and minima are no longer evenly distributed.

For a spacing of $d = \lambda/2$, the radiation pattern exhibits characteristics similar to an end-fire array, where maximum radiation occurs along the axis of the sources ($\theta = 0^\circ$ and 180°), while reduced radiation occurs in the perpendicular direction. However, due to unequal amplitudes, the lobes are not identical in magnitude, and the symmetry of the pattern is broken. The ratio of amplitudes (A_1/A_2) plays a significant role in controlling the depth of nulls and the level of side lobes.Thus, both amplitude and phase control are essential parameters in antenna arrays.

Source Code

```
# Two Isotropic Sources (Unequal Amplitude + Opposite Phase)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
```

```
lam = 1.0
d = lam / 2
k = 2 * np.pi / lam
# Amplitude weights
A1 = 1
A2 = 0.5   # unequal amplitude
# Angle range
theta = np.linspace(0, 2*np.pi, 1000)
# Array Factor
AF = A1 - A2 * np.exp(1j * k * d * np.cos(theta))
# Normalize
AF_norm = np.abs(AF) / np.max(np.abs(AF))

# Plot
plt.figure(figsize=(6,6))
plt.polar(theta, AF_norm)
plt.title("Far-Field Pattern\nUnequal Amplitude + Opposite Phase\n(d =   /2)")
plt.show()
```

Output

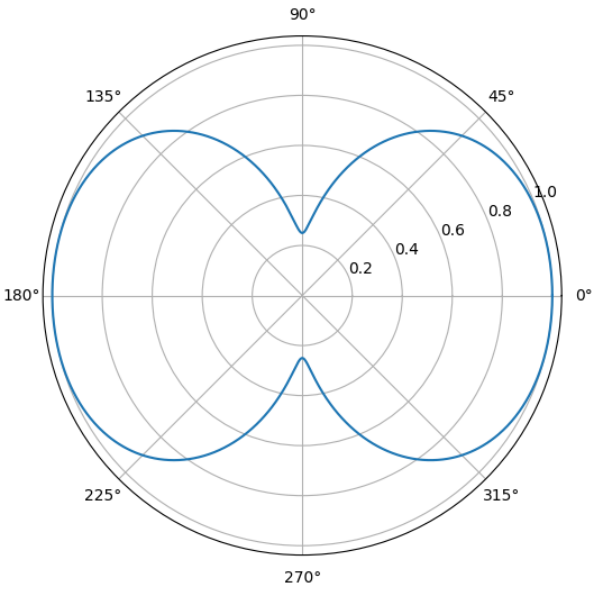


Figure 8: Far-Field Radiation Pattern (Unequal Amplitude and Opposite Phase)

Result and Discussion

The simulation results demonstrate the radiation pattern of two isotropic point sources with unequal amplitudes and opposite phase. Unlike the equal amplitude case, the pattern becomes asymmetric, and the nulls are no longer completely zero due to unequal contribution from the sources. The main lobes are slightly shifted, and the radiation intensity varies in different directions. This shows that amplitude weighting significantly influences the radiation pattern. By adjusting amplitudes, it is possible to control side lobes and improve antenna performance. The results are consistent with theoretical expectations and highlight the importance of amplitude control in antenna array design.

Problem No: 09
Problem Name: Study and Simulation of the 2D Radiation Pattern of a Broadside 5-Element Array Antenna

Objective

- To understand the behavior of a multi-element broadside array antenna.
- To study the effect of increasing the number of elements on radiation pattern.
- To analyze the beamwidth and side lobe formation.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A broadside array antenna consists of multiple identical elements arranged linearly with equal spacing and excited with equal amplitude and phase. In such an array, the maximum radiation occurs perpendicular to the array axis, known as the broadside direction ($\theta = 90°$).

For a uniform linear array with N elements and spacing d, the array factor is expressed as:

$$AF(\theta) = \left| \sum_{n=0}^{N-1} e^{jnkd\cos\theta} \right|$$

where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- d is the spacing between elements

In a broadside configuration, all elements are excited in phase ($\beta = 0$), ensuring constructive interference in the perpendicular direction.

When the number of elements increases (e.g., $N = 5$), the radiation pattern becomes more directional. The main lobe becomes narrower, indicating higher directivity and improved beam focusing. However, additional side lobes also appear, which represent undesired radiation in other directions.

The beamwidth of the main lobe is inversely proportional to the number of elements, meaning more elements produce a sharper beam. At the same time, side lobes may become more complex, requiring amplitude control (tapering) to suppress them.

Thus, increasing the number of elements in a broadside array improves directivity but also introduces challenges such as side lobe management.

Source Code

Step 1: Broadside 5-Element Array (2D Radiation Pattern)

```
# Broadside array antenna (5-element)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 5   # number of elements

# Angle range
theta = np.linspace(0, np.pi, 1000)

# Array Factor
```

```
AF = np.zeros_like(theta, dtype=complex)

for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

# Normalize
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

# Plot
plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)

ax.set_title("2D Radiation Pattern (5-Element Broadside Array)")
plt.show()
```

Output

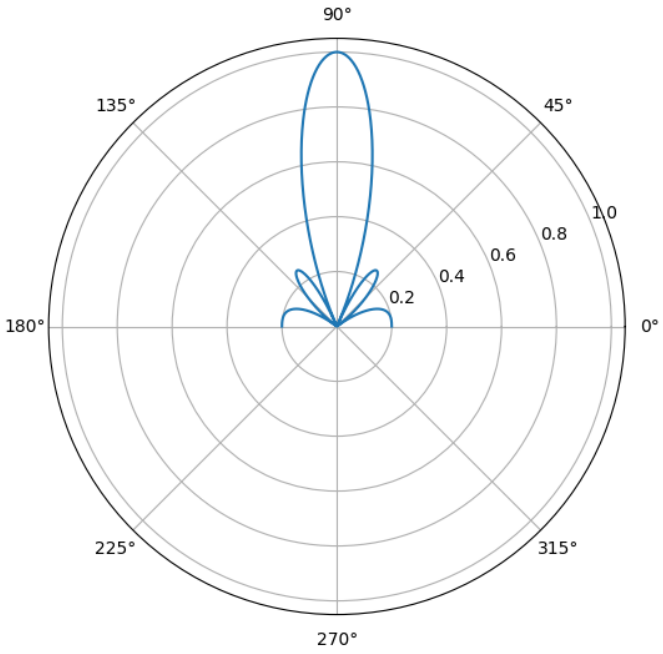


Figure 9: 2D Radiation Pattern of 5-Element Broadside Array Antenna

Result and Discussion

The simulation results show the radiation pattern of a 5-element broadside array antenna. Compared to the 3-element array, the main lobe becomes significantly narrower, indicating higher directivity and better beam focusing. At the same time, multiple side lobes appear on both sides of the main beam, which represent unwanted radiation. This demonstrates that increasing the number of elements enhances directional performance but also increases side lobe complexity. The results confirm that antenna arrays can effectively control radiation patterns, making them suitable for applications such as radar and wireless communication systems where high directivity is required.

Problem No: 10
Problem Name: Study and Simulation of the 2D Radiation Pattern of a Broadside 7-Element Array Antenna

Objective

- To study the behavior of a large antenna array system.
- To analyze the effect of increasing the number of elements on radiation pattern.
- To observe beam narrowing and side lobe formation.
- To simulate and visualize the 2D radiation pattern using Python.

Theory

A broadside array antenna is formed by arranging multiple identical antenna elements in a straight line with equal spacing and exciting them with equal amplitude and phase. In such arrays, maximum radiation occurs perpendicular to the array axis ($\theta = 90°$), which is known as the broadside direction.

The radiation pattern of a uniform linear array is determined by its array factor (AF), given by:

$$AF(\theta) = \left| \sum_{n=0}^{N-1} e^{jnkd\cos\theta} \right|$$

where:

- $k = \frac{2\pi}{\lambda}$ is the wave number
- d is the spacing between elements
- N is the number of elements

For a broadside array, all elements are excited in phase ($\beta = 0$), resulting in constructive interference in the perpendicular direction.

When the number of elements increases to $N = 7$, the radiation pattern becomes highly directional. The main lobe becomes very narrow, indicating a high degree of directivity and strong beam focusing. However, the number of side lobes also increases, and their spacing becomes smaller.

The beamwidth is inversely proportional to the number of elements, so increasing N reduces the beamwidth significantly. While this improves directivity, it also requires careful design to control side lobes using techniques such as amplitude tapering.

Thus, a 7-element broadside array demonstrates how increasing the number of antenna elements enhances performance but introduces additional complexity in radiation control.

Source Code

Step 1: Broadside 7-Element Array (2D Radiation Pattern)

```
# Broadside array antenna (7-element)
import numpy as np
import matplotlib.pyplot as plt

# Parameters
lam = 1
d = lam / 2
k = 2 * np.pi / lam
N = 7   # number of elements

# Angle range
theta = np.linspace(0, np.pi, 1000)
```

```
# Array Factor
AF = np.zeros_like(theta, dtype=complex)

for n in range(N):
    AF += np.exp(1j * n * k * d * np.cos(theta))

# Normalize
AF_mag = np.abs(AF)
AF_mag = AF_mag / np.max(AF_mag)

# Plot
plt.figure(figsize=(6,6))
ax = plt.subplot(111, projection='polar')
ax.plot(theta, AF_mag)

ax.set_title("2D Radiation Pattern (7-Element Broadside Array)")
plt.show()
```

Output

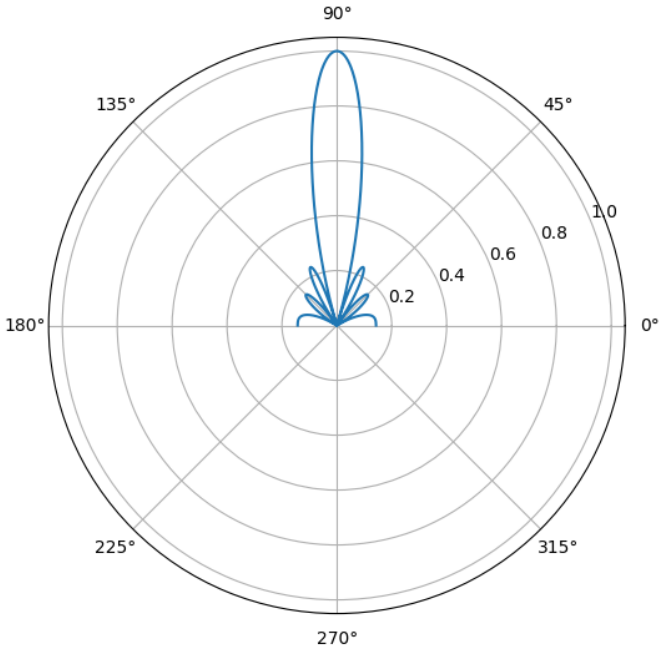


Figure 10: 2D Radiation Pattern of 7-Element Broadside Array Antenna

Result and Discussion

The simulation results demonstrate the radiation pattern of a 7-element broadside array antenna. Compared to arrays with fewer elements, the main lobe becomes extremely narrow, indicating a high level of directivity and precise beam control. Multiple side lobes are observed, and their number increases as the number of elements grows. This shows that while increasing the number of elements significantly improves directional performance, it also introduces more side lobes that may cause unwanted radiation. The results confirm that antenna arrays can effectively shape radiation patterns, and proper design techniques are required to balance directivity and side lobe suppression.